

Professional Summary

Data Scientist and Mathematical Engineer with experience in ML, NLP, and optimization, applying Python, R, SQL, and Azure to solve real business problems. Proven track record improving marketing ROI by 20% and cutting processing time by ~70% through scalable models and workflow automation. Skilled at communicating insights to both technical and non-technical stakeholders to support strategic decisions.

EDUCATION

Universidad EAFIT, Medellín

BSc in Mathematical Engineering | GPA: 4.30 / 5.00 | Graduated 2023

PROJECTS

A hybrid metaheuristic with a split-based procedure for the batch parallel machine scheduling Jul 2022 – Nov 2023 Python

- Implemented a hybrid metaheuristic combining Evolutionary Path Relinking and Variable Neighborhood Search to solve complex batch parallel machine scheduling problems.
- Achieved significantly improved schedule quality vs baseline heuristics..

Sentiment Analysis of Tweets in Colombia Jan 2022-May 2022 Python

- Collected and preprocessed large-scale Twitter data and applied word embeddings and emotion-detection models to analyze public sentiment during the pandemic.
- Identified key emotional trends and time periods, generating insights relevant for public communication strategies.

Neural Networks for Image Recognition Jul 2023-Nov 2023 Python

- Trained and fine-tuned convolutional neural networks for image classification on diverse datasets using PyTorch and TensorFlow.
- Improved baseline accuracy by 7% through architecture tuning, regularization, and data augmentation.

EXPERIENCE

Data Scientist, Vensure Employer Solutions Dec 2024 – Present

- Built a Teams profile-photo classifier (professional vs non-professional) on a small internal dataset, achieving 76% accuracy through neural network tuning and facial recognition.
- Designed a news alert mapping system using OCR and NER, reducing processing time by ~70% via parallelization and code optimization, and enabling timely geo-tagged alerts for global leaders.
- Automated performance-monitoring pipelines by integrating Azure databases with Python, triggering email alerts for underperformance and accelerating management responses from 2 weeks to 5 days.

Data Analyst, Ziruma May 2024 – Nov 2024

- Analyzed marketing and advertising data using Python and SQL, optimizing campaigns and improving key performance metrics by 15%.
- Built and deployed ML models to predict customer behavior and improve ad targeting efficiency.
- Developed automated KPI dashboards and tracking scripts, saving 10+ hours per week and enabling faster performance reviews using Matplotlib and Tableau.
- Partnered with cross-functional teams to integrate data-driven recommendations into marketing strategy, increasing ROI by 20% through optimized targeting.

Data Scientist (Internship), Accenture Jan 2023 - Jul 2023

- Developed optimization models in Python and R to streamline water-usage processes for a major Colombian oil company, reducing resource waste and improving operational efficiency.
- Built ML models (including neural networks) to predict operational outcomes, supporting data-driven decisions on resource allocation.
- Led data analysis for a global initiative supporting unemployed youth in Colombia, delivering reports with insights that shaped program design and stakeholder strategy.
- Used Python, SQL, and visualization tools to prepare detailed reports for international banking clients, strengthening data-driven decision-making.

Skills & Competencies

- Programming & Data:** Python (advanced – pandas, scikit-learn), R, SQL, MATLAB
- ML & AI:** Neural networks, deep learning, NLP (NER, sentiment analysis), OCR, optimization, statistical modeling
- Cloud & Tools:** Azure, Power BI, Tableau, Git

Languages

- Spanish:** Native
- English:** Advanced (IELTS 7.0, 2023)

ACHIEVEMENTS AND RECOGNITIONS

Sapiencia Best Student Award

- Top-performing student among Medellín public institutions based on national ICES exam scores.